CLASS: JSS 1 - 2

INTRODUCTION TO WEB DEVELOPMENT

Web development involves creating websites and web applications that can be accessed on the internet. A website is a collection of web pages that provide information, services, or tools to users.

THERE ARE TWO PRIMARY PARTS OF WEB DEVELOPMENT:

1. **Frontend (Client-side):** This involves creating the visual part of a website that users interact with using HTML, CSS, and JavaScript.
2. **Backend (Server-side):** This involves creating the server and database that manage the website's data and functionality.

For this course, we will focus on **Frontend Development** using **HTML (HyperText Markup Language)** and **CSS (Cascading Style Sheets)**.

UNDERSTANDING HTML

HTML stands for **HyperText Markup Language**. It is the standard language used to create web pages.

KEY CONCEPTS OF HTML:

- **Elements:** HTML is made up of elements enclosed in tags.
- **Tags:** Tags are used to structure the content on a web page.
- **Attributes:** Attributes provide additional information about an element (e.g., the src attribute for an image).

Example of a basic HTML element:

This is a paragraph.

SETTING UP A SIMPLE HTML FILE

To create a simple HTML file, follow these steps:

Steps to Create an HTML File:

1. Open a text editor (like Notepad or VS Code).
2. Save the file with a .html extension.
3. Write the basic HTML structure.

Basic HTML Structure:

```
<!DOCTYPE html>
<html>
<head>
  <title>My First Webpage</title>
</head>
<body>
  <h1>Welcome to My Website</h1>
  <p>This is my first HTML page!</p>
</body>
</html>
```

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USING HEADINGS AND PARAGRAPHS

Headings are used to create titles and subtitles on a webpage. Paragraphs are used to add text content.

Headings:

There are six heading levels in HTML: <h1> to <h6>.

<h1>This is a main heading</h1>

<h2>This is a subheading</h2>

<h3>This is a smaller subheading</h3>

Paragraphs:

Paragraphs are added using the tag.

This is a paragraph of text.

ADDING TEXT FORMATTING TAGS

HTML provides tags to format text in different ways.

Common Text Formatting Tags:

This text is bold.

This text is italic.

<u>This text is underlined.</u>

 <!-- Line break -->

This text is important.

This text is emphasized.

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CREATING LISTS IN HTML

HTML allows you to create both ordered and unordered lists.

Unordered List:

An unordered list uses bullets.

```
<ul>
  <li>Item 1</li>
  <li>Item 2</li>
  <li>Item 3</li>
</ul>
```

ORDERED LIST:

An ordered list uses numbers.

```
<ol>
  <li>First item</li>
  <li>Second item</li>
  <li>Third item</li>
</ol>
```

ADDING LINKS USING <A> TAG

The <a> tag is used to add hyperlinks to a webpage.

Example of a Link:

```
<a href="https://www.example.com">Visit Example</a>
```

- **href:** Specifies the URL of the link.

Opening Links in a New Tab:

```
<a href="https://www.example.com" target="_blank">Visit Example</a>
```

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ADDING IMAGES USING <IMG> TAG

The <img> tag is used to add images to a webpage.

Example of an Image:

```
<img src="image.jpg" alt="Description of the image">
```

- **src:** Specifies the path to the image.
- **alt:** Provides alternative text if the image cannot be displayed.

CREATING TABLES USING <TABLE> TAG

The <table> tag is used to create tables in HTML.

Example of a Simple Table:

```
<table border="1">
  <tr>
    <th>Name</th>
    <th>Age</th>
  </tr>
  <tr>
    <td>John</td>
    <td>25</td>
  </tr>
  <tr>
    <td>Jane</td>
    <td>22</td>
  </tr>
</table>
```

- **<table>:** Creates the table.



- **<tr>:** Defines a row.
- **<th>:** Defines a table header.
- **<td>:** Defines a table cell.

USING FORMS IN HTML

Forms are used to collect user input.

Example of a Form:

```
<form action="submit.php" method="post">
  <label for="name">Name:</label>
  <input type="text" id="name" name="name">
  <br>
  <label for="email">Email:</label>
  <input type="email" id="email" name="email">
  <br>
  <input type="submit" value="Submit">
</form>
```

- **<form>:** Creates the form.
- **<input>:** Creates input fields.
- **<label>:** Describes the input fields.
- **action:** Specifies where to send the form data.
- **method:** Specifies the HTTP method to use (GET or POST).

STRUCTURING A BASIC WEBPAGE

A well-structured webpage should include a header, main content, and footer sections.

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Example of a Structured Webpage:

```
<!DOCTYPE html>
<html>
<head>
  <title>My Website</title>
</head>
<body>
  <!-- Header Section -->
  <header>
    <h1>Welcome to My Website</h1>
  </header>

 <!-- Main Content Section -->
  <main>
    <p>This is where the main content goes.</p>
    <a href="#">Read more...</a>
  </main>

  <!-- Footer Section -->
  <footer>
    <p>Copyright &copy; 2025. All rights reserved. </p>
  </footer>
</body>
</html>
```

This structure ensures that your webpage is organized and easy to navigate.



Mini Project: Build a Personal Webpage

```
<!DOCTYPE html>
<html>
<head>
  <title>My Website</title>
</head>
<body>
<!-- About Me Section -->
<section id="about">
  <h2>About Me</h2>
  <p>I am a student learning web development. I enjoy creating webpages and learning new technologies.</p>
</section>

<!-- Hobbies Section -->
<section id="hobbies">
  <h2>My Hobbies</h2>
  <ul>
    <li>Reading books</li>
    <li>Playing video games</li>
    <li>Learning to code</li>
    <li>Listening to music</li>
  </ul>
</section>
```



```
<!-- Contact Section -->
<section id="contact">
  <h2>Contact Me</h2>
  <p>You can reach me at:</p>
  <ul>
    <li>Email: myemail@example.com</li>
    <li>Phone: +123 456 7890</li>
  </ul>
</section>

<!-- Footer Section -->
<footer>
  <p>&copy; 2025 [Your Name]. All rights reserved. </p>
</footer>

</body>
</html >
```

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